

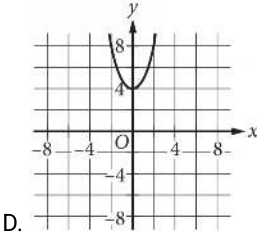
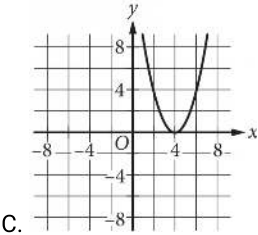
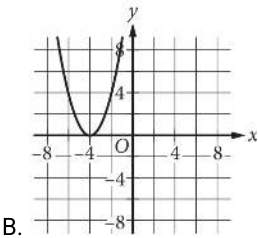
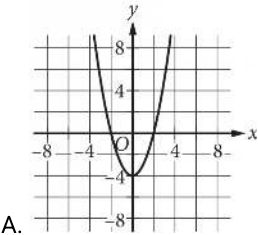
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

$$f(x) = x^2 + 4$$

The function f is defined as shown. Which of the following graphs in the xy -plane could be the graph of $y = f(x)$?

Answer



Rationale

Choice D is correct. For the quadratic function $f(x) = x^2 + 4$, the vertex of the graph is $(0, 4)$, and because the x^2 term is positive, the vertex is the minimum of the function. Choice D is the only option that meets these conditions.

Choices A, B, and C are incorrect. The vertex of each of these graphs doesn't correspond to the minimum of the given function.